HEXIN CHEN

Beijing, P.R.China
https://github.com/JJsoga | hx-chen24@mails.tsinghua.edu.cn

EDUCATION

Tsinghua University, China *09/2024–06/2028*
B.S. in Mathematics and Physics; B.E. / B.S. in Software Engineering (Dual Degree)
GPA: 3.8/4.0 (Ranking Top 15%)
Coursework: Calculus, Linear Algebra, Probability and Stochastic Progress, Discrete Mathematics I-II, Data Structures, Fundamental Programming, Principles of Database Systems, Computer Organization Principle

RESEARCH EXPERIENCE

Wearable Breathing Sensing for Human-Centered Computing
Research Assistant; University of Chicago *06/2026–09/2026*
• Contributed to a research project exploring cognitive-load sensing through breathing signals captured by contact microphones integrated into an eyewear form factor.
• Developed signal-processing and machine-learning components for extracting physiological features and estimating cognitive load from wearable sensor data.
• Investigated user variability and practical challenges in physiological sensing, including environmental interference and differences in breathing patterns.
• Conducted model comparisons, cross-participant evaluation, and controlled experiments to assess prediction reliability and robustness.
•Implemented a real-time sensing and visualization prototype, gaining experience in end-to-end system development, experimental evaluation, and human-centered computing research.
Publication: Manuscript submitted to CHI with Prof. Pedro Lopes during my research internship.

Audio Signal Representation Project
Research Assistant; Advisor: Dr. Jiangyan Yi, Tsinghua University *06/2025 - 03/2026*
• Worked on a neural audio representation pipeline for academic research and prototype-level experiments.
• Refactored the quantization module to improve modularity, reconstruction stability, and downstream usability.
• Implemented and evaluated alternative vector-quantization designs for separating global audio patterns from local acoustic details.
• Ran controlled experiments and ablation studies to compare system components.

Motion-Robust IMU Signal Processing for Wearable Interaction
Research Assistant; Tsinghua University *09/2024–07/2025*
• Analyzed multi-axis accelerometer signals in both time and frequency domains, identifying noise patterns and non-stationary motion artifacts in fine-grained wearable sensing data.
• Developed an adaptive signal filtering pipeline to reduce motion-induced jitter while preserving subtle intentional movement cues under low-latency constraints.
• Improved trajectory stability and repeatability in prototype experiments with minimal impact on system responsiveness.

LEADERSHIP & ACTIVITIES

Team Captain, "Qinglai Zhixing" Social Practice Team, Tsinghua University *07/2025–08/2025*
Team Lead, Tsinghua–CUC Game Jam *12/2024*
Group Lead, Student Practice Committee, Wuqiong College, Tsinghua University *08/2025–09/2025*
Contestant, MCM/ICM Mathematical Modeling Contest *02/2025*

HONORS & AWARDS

Academic Excellence Scholarship *10/2025*
Merit Scholarship *10/2025*
Silver Award, Summer Social Practice Team, Tsinghua University *12/2025*
Research Innovation Scholarship *11/2026*
Xitai Scholarship *11/2026*

CORE SKILLS

Languages: Mandarin Chinese (Native); English (TOEFL 100)

Mathematics: Calculus, Linear Algebra, Probability & Stochastic Processes, Discrete Mathematics, Mathematical Modeling, Statistical Analysis

Programming & Engineering: Python, C++, C, Java, Arduino; Signal Processing, Databases, Computer Systems, Software Engineering

Machine Learning & Research: Machine Learning, Time-Series Analysis, Feature Engineering, Model Training & Evaluation, Controlled Experiments, Statistical Analysis

Leadership & Collaboration: Team Leadership, Cross-functional Collaboration, Project Coordination, External Outreach